Problem Solving II: Data and Uncertainty

For some entertaining general discussion, see chapters I-5 and I-6 of the Feynman lectures. There is a total of **73** points.

1 Basic Probability

Idea 1

If a quantity X has the probability distribution $p(x)$, that means

$$\text{the probability that } a \le X \le b \text{ is } \int_a^b p(x)\,dx.$$

In particular, the total probability has to sum to one, so

$$\int_{-\infty}^{\infty} p(x)\,dx = 1.$$

Using the probability distribution, we can calculate expectation values, i.e. averages. For example, the expectation value of X, also called the mean, is

$$\langle X \rangle = \int_{-\infty}^{\infty} xp(x)\,dx$$

while the expectation value of an arbitrary function of X is

$$\langle f(X) \rangle = \int_{-\infty}^{\infty} f(x)p(x)\,dx.$$

One especially important quantity is the variance of X, defined as

$$\operatorname{var} X = \langle X^2 \rangle - \langle X \rangle^2.$$

The standard deviation is defined by $\sigma_X = \sqrt{\operatorname{var} X}$. It describes how "spread out" the distribution of X is, and it will play an important role in uncertainty analysis.

[1] **Problem 1.** Suppose that x is a length. What are the dimensions of $p(x)$, $\langle X \rangle$, $\operatorname{var} X$, and σ_X?

Solution. Since $p(x)\,dx$ is dimensionless, we have

$$[p(x)] = L^{-1}$$

where L denotes length. Similarly,

$$[\langle X \rangle] = L, \quad [\operatorname{var} X] = L^2, \quad [\sigma_X] = L.$$

Example 1

Trains arrive at a train station every 10 minutes. If I arrive at a random time, and X is the number of minutes I have to wait, what is the standard deviation of X?

Solution

We see that X can be anywhere between 0 and 10, with all possibilities equally likely, so

$$p(x) = \begin{cases} 1/10 & 0 \le x \le 10, \\ 0 & \text{otherwise} \end{cases}$$

where the denominator guarantees the total probability is 1. We have

$$\langle X \rangle = \int_{-\infty}^{\infty} xp(x)\,dx = \int_0^{10} \frac{x}{10}\,dx = 5$$

which makes sense, as I should have to wait half the maximum time on average, and

$$\langle X^2 \rangle = \int_{-\infty}^{\infty} x^2 p(x)\,dx = \int_0^{10} \frac{x^2}{10}\,dx = \frac{100}{3}.$$

Then the standard deviation is

$$\sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2} = \frac{5}{\sqrt{3}}\,\text{min}.$$

[3] **Problem 2.** Consider an exponentially distributed quantity,

$$p(x) = \begin{cases} ae^{-ax} & x \ge 0, \\ 0 & \text{otherwise}, \end{cases}$$

for some $a > 0$. Verify that the total probability is 1, and compute the mean and standard deviation.

Solution. First, to check normalization,

$$\int_{-\infty}^{\infty} p(x)dx = \int_0^{\infty} ae^{-ax}dx = \int_0^{\infty} e^{-u}du = 1 - 0 = 1.$$

Now, the mean can be evaluated using integration by parts,

$$\langle x \rangle = \int_0^{\infty} xae^{-ax}\,dx = -xe^{-ax}\Big|_0^{\infty} + \int_0^{\infty} e^{-ax}\,dx = 0 - \frac{1}{a}e^{-ax}\Big|_0^{\infty} = \frac{1}{a}.$$

To calculate the standard deviation, we must evaluate

$$\langle x^2 \rangle = \int_0^{\infty} x^2 ae^{-ax}dx = 0 + \int_0^{\infty} (2x)e^{-ax}dx = \frac{2}{a}\langle x \rangle = \frac{2}{a^2}.$$

We thus conclude

$$\sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2} = \sqrt{\frac{2}{a^2} - \frac{1}{a^2}} = \frac{1}{a}.$$

[2] **Problem 3.** The purpose of subtracting $\langle X \rangle^2$ in the variance is to make sure it doesn't change when a constant is added to x, since shifting something left or right on the number line shouldn't change its spread. Verify that for any constant c, $\text{var}\,X = \text{var}(X + c)$.

Solution. We have

$$\operatorname{var}(X+c) = \langle (X+c)^2 \rangle - \langle X+c \rangle^2.$$

By the definition of the expectation value, we have

$$\langle A+B \rangle = \langle A \rangle + \langle B \rangle, \quad \langle cA \rangle = c\langle A \rangle$$

for any quantities A and B and any constant c. Thus,

$$\operatorname{var}(X+c) = \langle X^2 \rangle + \langle 2Xc \rangle + \langle c^2 \rangle - \langle X \rangle^2 - 2\langle X \rangle \langle c \rangle - \langle c \rangle^2 = \operatorname{var} X$$

as desired.

[3] **Problem 4.** We say X is normally distributed if

$$p(x) \propto e^{-a(x-b)^2}$$

for some $a > 0$. For simplicity, let's shift X so that it's centered about $x = 0$, so

$$p(x) \propto e^{-ax^2}.$$

You may use the result given in **P1**,

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$$

Find the constant of proportionality in $p(x)$, the mean, and the standard deviation.

Solution. Let $p(x) = ke^{-ax^2}$. We fix the constant k by demanding normalization,

$$\int_{-\infty}^{\infty} ke^{-ax^2}\, dx = \int_{-\infty}^{\infty} \frac{k}{\sqrt{a}} e^{-u^2} du = 1.$$

Using the provided integral, we conclude

$$k = \sqrt{\frac{a}{\pi}}.$$

The mean is clearly zero, since the distribution is symmetric about that point. Thus, we have

$$\operatorname{var} X = \langle X^2 \rangle = \sqrt{\frac{a}{\pi}} \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{1}{a\sqrt{\pi}} \int_{-\infty}^{\infty} u^2 e^{-u^2} du.$$

This remaining integral can be evaluated using integration by parts,

$$\int_{-\infty}^{\infty} u^2 e^{-u^2}\, du = -\frac{1}{2} u e^{-u^2} \Big|_{-\infty}^{\infty} + \frac{1}{2} \int_{-\infty}^{\infty} e^{-u^2}\, du = 0 + \frac{\sqrt{\pi}}{2}$$

from which we conclude

$$\operatorname{var} X = \frac{1}{a\sqrt{\pi}} \frac{\sqrt{\pi}}{2}, \quad \sigma = \frac{1}{\sqrt{2a}}.$$

Remark

As you saw in problem 4, normal distributions are characterized entirely by their mean and standard deviation. Moreover, the central limit theorem roughly states that the sum of many independent random variables is approximately normally distributed. As a result, final experimental results are often approximately normally distributed, which means we can describe them with only the mean and standard deviation. That is the reason why the next part of this problem set will focus on calculating standard deviations (i.e. uncertainties).

[2] **Problem 5.** If two random variables X_1 and X_2 are independent, then

$$\langle X_1 X_2 \rangle = \langle X_1 \rangle \langle X_2 \rangle.$$

Use this result to show that

$$\text{var}(X_1 + X_2) = \text{var}(X_1) + \text{var}(X_2)$$

which implies that the standard deviation "adds in quadrature",

$$\sigma_{X_1+X_2} = \sqrt{\sigma_{X_1}^2 + \sigma_{X_2}^2}.$$

This is an important result we'll use many times below.

Solution. By definition, we have

$$\text{var}(X_1 + X_2) = \langle (X_1 + X_2)^2 \rangle - \langle X_1 + X_2 \rangle^2$$

Using the properties listed in problem 3,

$$\begin{aligned} \text{var}(X_1 + X_2) &= \langle X_1^2 \rangle + 2\langle X_1 X_2 \rangle + \langle X_2^2 \rangle - \langle X_1 \rangle^2 - 2\langle X_1 \rangle \langle X_2 \rangle - \langle X_2 \rangle^2 \\ &= \text{var}(X_1) + \text{var}(X_2) + 2(\langle X_1 X_2 \rangle - \langle X_1 \rangle \langle X_2 \rangle) \end{aligned}$$

When X_1 and X_2 are independent, the last term vanishes, giving

$$\text{var}(X_1 + X_2) = \text{var}(X_1) + \text{var}(X_2).$$

2 Uncertainty Propagation

Idea 2

When a physical quantity is measured in an experiment and reported as $x \pm \Delta x$, it is uncertain what the true value of the quantity is. If the quantity has a probability distribution $p(x)$, then the reported uncertainty Δx is essentially the standard deviation of $p(x)$.

Remark

You can use intuition and experience to assign uncertainties to real measurements. For example, if you're using a clock that times only to the nearest second, you might take $\Delta t = 0.5\,\text{s}$. If you're using a good ruler, which has millimeter markings, you might take

$\Delta x = 0.5\,\text{mm}$. Of course, the ultimate test is the results: if you assigned the uncertainties right, your final uncertainty should encompass the true result most (but not all) of the time.

[1] **Problem 6.** Suppose x has uncertainty Δx and y has uncertainty Δy, where x and y are independent. Explain why the uncertainty of $x + y$ is

$$\Delta(x+y) = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

This is called "addition in quadrature". What is the uncertainty of $x - y$? How about $x + x$?

Solution. For independent variables, $\text{var}(X_1 + X_2) = \text{var}(X_1) + \text{var}(X_2)$. Since our uncertainties represent the standard deviation, $\sigma_X = \sqrt{\text{var}(X)}$, we have

$$\Delta(x+y) = \sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

Since $x - y = x + (-y)$ and $\Delta(-y) = \Delta y$, we get that $\Delta(x-y) = \Delta(x+y)$. Finally, by linearity we clearly have $\Delta(x+x) = 2\Delta x$. (The formula above doesn't apply, because x isn't independent of x.)

Remark

Note how this differs from "high school" uncertainty analysis. In school, you might be told to show uncertainty using significant figures, and when adding two things, to keep only the figures that are significant in both of them. That corresponds to

$$\Delta(x+y) = \max(\Delta x, \Delta y)$$

which is an underestimate. Or, you might be told that the uncertainty needs to encapsulate all the possible values, which implies that

$$\Delta(x+y) = \Delta x + \Delta y$$

which is an overestimate, since the errors could cancel.

Example 2: $F = ma$ 2016 25

Three students make measurements of the length of a 1.50 m rod. Each reports an uncertainty estimate representing an independent random error applicable to the measurement.

- Alice performs a single measurement using a 2.0 m tape measure, to within 2 mm.
- Bob performs two measurements using a wooden meter stick, each to within 2 mm, which he adds together.
- Christina performs two measurements using a machinist's meter rule, each to within 1 mm, which she adds together.

Rank the measurements in order of their uncertainty.

Solution

The uncertainty in Alice's measurement is 2 mm. The uncertainty in Bob's is $2\sqrt{2}$ mm by quadrature, while the uncertainty in Christina's is $\sqrt{2}$ mm by quadrature. So the lowest uncertainty is Christina's, followed by Alice's, followed by Bob's.

[1] **Problem 7.** Given N independent measurements of the same quantity with the same uncertainty, $x_i \pm \Delta x$, find the uncertainty of their sum. Hence show the uncertainty of their average is $\Delta x/\sqrt{N}$.

This result is extremely important, since repeating trials is one of the main ways to reduce uncertainty. But it's important to remember that the results derived above hold only for independent measurements. For example, taking a single measurement, then averaging that single number with itself 100 times certainly wouldn't reduce the uncertainty at all!

Solution. The uncertainty of their sum ΔX can be found by adding in quadrature,

$$\Delta X = \sqrt{\sum_{i=1}^{N} (\Delta x_i)^2} = \sqrt{N}\Delta x.$$

Therefore, the uncertainty of the average is

$$\frac{\Delta X}{N} = \frac{\Delta x}{\sqrt{N}}.$$

Idea 3: Tangent Line Approximation

If x has uncertainty Δx, and $f(x)$ can be approximated by its tangent line,

$$f(x') \approx f(x) + (x' - x)f'(x),$$

within the region $x \pm \Delta x$, then the distribution of $f(x)$ is just a shifted and stretched version of the distribution of x. Moreover, $f(x)$ has approximate uncertainty $|f'(x)|\,\Delta x$.

[1] **Problem 8.** If x is positive and has uncertainty Δx, find the uncertainties of x^2, $\sqrt{x}$, $1/x$, $\log x$, and e^x using the tangent line approximation.

Solution. Using idea 3, we simply read off the answers,

$$\Delta(x^2) = 2x\Delta x \qquad \Delta(\sqrt{x}) = \frac{\Delta x}{2\sqrt{x}} \qquad \Delta(1/x) = \frac{\Delta x}{x^2}$$

$$\Delta(\log(x)) = \frac{\Delta x}{x} \qquad \Delta(e^x) = e^x \Delta x$$

[1] **Problem 9.** The tangent line approximation doesn't always make sense. Suppose x is measured as $0 \pm \Delta x$. Show that the results of problem 8 for the uncertainties of x^2 and $\sqrt{x}$ give nonsensical results. In each case, what would be a more reasonable uncertainty to report?

Solution. Naively applying the formulas above would give uncertainties of zero and infinity, which are clearly unphysical. In these cases, reasonable guesses for the uncertainty would be $(\Delta x)^2$ and $\sqrt{\Delta x}$, since by dimensional analysis, they give the typical spread of the distributions of x^2 and $\sqrt{x}$.

Still, there is something more subtle going on here. In cases like these, applying the transformation distorts the probability distribution, beyond just a shifting and stretching, so it doesn't stay normally distributed. In Olympiads, this doesn't matter since we only want rough estimates, but the proper way to handle this would be to describe the full probability distribution. However, this often can't be done analytically.

When professional physicists run into situations like these, they often use Monte Carlo: they numerically compute many values, starting with randomly drawn inputs, and use that to infer the probability distribution. On Olympiads, you should just fall back to something reasonable.

[2] **Problem 10.** Consider two independent, positive quantities with values and uncertainties $x \pm \Delta x$ and $y \pm \Delta y$.

(a) By writing xy as $\exp(\log x + \log y)$, show that the uncertainty of xy is

$$\Delta(xy) = xy\sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}.$$

(b) If we set $x = y$, then we find

$$\Delta(x^2) = x^2\sqrt{2\left(\frac{\Delta x}{x}\right)^2} = \sqrt{2}\, x\Delta x.$$

On the other hand, in problem 8 we found $\Delta(x^2) = 2x\Delta x$. Which result is correct?

(c) Find the uncertainty of x/y.

Solution. (a) We can write

$$xy = \exp(\log x + \log y)$$

which implies

$$\Delta(xy) = \exp(\log x + \log y)\Delta(\log x + \log y) = xy\sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}.$$

(b) The result $\Delta(x^2) = 2x\Delta x$ is correct, since the formula for $\Delta(xy)$ assumes x and y are independent, which fails when we set $y = x$.

(c) We have

$$\frac{x}{y} = \exp(\log x - \log y)$$

and by a very similar calculation to part (a), we conclude

$$\Delta(x/y) = \frac{x}{y}\sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}.$$

[2] **Problem 11.** A student launches a projectile with speed $v = 5 \pm 0.1\,\text{m/s}$ in gravitational acceleration $g = 9.81 \pm 0.01\,\text{m/s}^2$. The resulting range is $d = 1.5 \pm 0.02\,\text{m}$. Given that the launch angle was less than $45°$, find the launch angle, with uncertainty, assuming all uncertainties are independent.

Solution. From the projectile range equation $d = v^2 \sin(2\theta)/g$, we get

$$\theta = \frac{1}{2} \arcsin\left(\frac{dg}{v^2}\right) = 18.03^\circ.$$

Now write $\sin(2\theta) = gd/v^2$. By the results of problem 10, the uncertainty of the right-hand side is

$$\frac{dg}{v^2}\sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta g}{g}\right)^2 + \left(\frac{2\Delta v}{v}\right)^2} = 0.0248.$$

The uncertainty of the left-hand side is $2\cos(2\theta)\Delta\theta$ by the tangent line approximation, so we have $\Delta\theta = 0.015$ rad $= 0.9^\circ$. Thus, the final result should be written as

$$\theta = 18.0^\circ \pm 0.9^\circ$$

where we removed a superfluous significant figure.

[2] **Problem 12.** Two physical quantities are related by $y = xe^x$.

(a) If x is measured to be 1.0 ± 0.1, find the resulting value of y, with uncertainty.

(b) If y is measured to be 2.0 ± 0.1, find the resulting value of x, with uncertainty.

Solution. (a) To find the central value of y, we plug in to get $y = e = 2.7183$. To find the error, we use the tangent line approximation,

$$\frac{dy}{dx} = e^x(x+1)$$

which gives us

$$\Delta y \approx e^x(x+1)\Delta x = 0.54.$$

Thus, rounding to a reasonable number of significant figures, we have

$$y = 2.7 \pm 0.5.$$

Note that it would be incorrect to apply the "addition in quadrature" rule for products,

$$\Delta y = xe^x\sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta(e^x)}{e^x}\right)^2}$$

because x and e^x aren't independent.

(b) To find the central value of x, we solve the equation $2 = xe^x$ numerically. This can be done using the method of iteration introduced in **P1**. That is, we have $x = 2e^{-x}$, so by repeatedly plugging $2e^{-\text{Ans}}$ into the calculator, we get $x = 0.8526$.

Under the tangent line approximation,

$$\Delta x \approx \frac{\Delta y}{e^x(x+1)} = 0.023.$$

Rounding to a reasonable number of significant figures, we conclude

$$x = 0.85 \pm 0.02.$$

Idea 4

For practical computations, it is often useful to use relative uncertainties. The relative uncertainty of x is $\Delta x/x$, and can be expressed as a percentage.

[1] **Problem 13.** Some basic relative uncertainty results.

(a) Show that the relative uncertainty of the product or quotient of two independent quantities is the square root of the sum of the squares of their relative uncertainties.

(b) Show that averaging the results of N independent trials as in problem 7 reduces the relative uncertainty by a factor of $\sqrt{N}$.

Solution. (a) Above we found that

$$\Delta(xy) = xy\sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

Dividing both sides by xy gives

$$\frac{\Delta(xy)}{xy} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2}$$

which is the desired result.

(b) Adding up the trials makes the absolute uncertainties add in quadrature, giving a factor of $\sqrt{N}$, while the total goes up by a factor of N, so the relative uncertainty goes down by a factor of $\sqrt{N}$. Dividing by N to form the average doesn't affect the relative uncertainty.

Remark

There are many situations where the rules above can't be used. For example, consider the uncertainty of $x + y^2/x$, where x and y are independent. You can calculate the uncertainty of either term with the standard rules, but you can't calculate the uncertainty of their sum, because the terms are not independent (both contain x).

In these cases, you can use the multivariable equivalent of the tangent line approximation,

$$f(x', y') \approx f(x, y) + (x' - x)\frac{\partial f}{\partial x} + (y' - y)\frac{\partial f}{\partial y}.$$

For independent x and y, adding the two contributions to the uncertainty in quadrature gives

$$\Delta f = \sqrt{\left(\frac{\partial f}{\partial x}\,\Delta x\right)^2 + \left(\frac{\partial f}{\partial y}\,\Delta y\right)^2}.$$

This general rule can also be used to quickly rederive the results of problem 10. However, it shouldn't be necessary in Olympiad problems. If you run into such situations in an experiment, often one of the uncertainties is much smaller, and can be neglected entirely.

3 Using Uncertainties

Example 3: $F = ma$ 2022 B21

Alice and Bob are given a long, thin rectangle of sheet metal. (It has been machined very precisely, so they can assume it is perfectly rectangular.) Using calipers, Alice measures the width of the rectangle as 1 cm with 1% uncertainty. Using a tape measure, Bob independently measures its length as 100 cm with 0.1% uncertainty. What are the relative uncertainties they should report for the area and the perimeter of the rectangle?

Solution

To compute the area, we multiply the two measurements, which means we add the relative uncertainties in quadrature,

$$\frac{\Delta A}{A} = \sqrt{(1\%)^2 + (0.1\%)^2} \approx 1\%.$$

Note that in this case, the relative uncertainty of Bob's measurement is negligible; the relative uncertainty of the area is approximately the relative uncertainty of Alice's measurement.

Computing the perimeter involves adding the measurements, which means the absolute uncertainties are added in quadrature instead. These are 0.01 cm and 0.1 cm for Alice and Bob's measurements, respectively, so the absolute uncertainty of Alice's measurement is negligible. Thus, the relative uncertainty of the perimeter is approximately the relative uncertainty of Bob's measurement, 0.1%.

In simple Olympiad experiments, often only one uncertainty will really matter. This can dramatically simplify calculations, but it might take a little thought to tell which one.

[3] **Problem 14.** As a speed test, solve $F = ma$ 2018 problems A12, A25, B19, and B25, and $F = ma$ 2019 problems A16, B18, and B25. Don't bother writing solutions.

[2] **Problem 15.** Suppose the goal of an experiment is to measure the ratio T_1/T_2 of the durations of two processes, where T_1 is about 15 seconds, and T_2 is about 3 seconds. Also suppose your stopwatch is only accurate to the nearest second. You have two minutes to perform measurements. Assume each measurement is independent.

(a) Do you think it's better to spend more total time measuring T_1, more total time measuring T_2, or an equal amount of time on both?

(b) To confirm this, find the fraction of time x spent measuring T_1 which minimizes the final uncertainty.

Calculations of this sort are common when doing Olympiad experimental physics. We will discuss these issues in more detail in the handout on experimental physics.

Solution. (a) Since T_2 is smaller, it takes less time to measure, and a single measurement of T_2 has a much higher relative uncertainty. So we definitely want more distinct measurements of T_2 than of T_1.

As for the total measurement time, note that if we split it equally, the uncertainty due to T_2 will still dominate, because $1/(3\sqrt{5}) > 1/15$. So we should spend more time measuring T_2.

(On the other hand, we shouldn't take a ratio as extreme as 5 to 1, because the benefit of additional measurements hits diminishing returns.)

(b) Explicitly, if ΔT is the absolute uncertainty on each measurement, then

$$\Delta(T_1/T_2) = \frac{T_1}{T_2}\sqrt{\left(\frac{\Delta T}{T_1\sqrt{N_1}}\right)^2 + \left(\frac{\Delta T}{T_2\sqrt{N_2}}\right)^2}$$

The total time T_t is constant, $N_1T_1 + N_2T_2 = T_t$ where $N_1T_1/T_t = x$. We want to minimize

$$f(x) = \frac{1}{T_1 x} + \frac{1}{(1-x)T_2}.$$

The derivative is

$$f'(x) = -\frac{1}{T_1x^2} + \frac{1}{T_2(1-x)^2}$$

and setting this to zero gives

$$x^2(1 - T_1/T_2) - 2x + 1 = 0.$$

The physical root is the one satisfying $0 < x < 1$, giving

$$x = \frac{1 - \sqrt{T_1/T_2}}{1 - T_1/T_2} = \frac{1}{1 + \sqrt{T_1/T_2}} \approx 0.3.$$

As expected from the arguments in part (a), this is more than 1/6 but less than 1/2.

[3] **Problem 16.** You can estimate g by finding the time needed for an object to roll down a ramp, with everything again made of household materials.

(a) Before starting, think about what the dominant sources of uncertainty will be, and how you can design the experiment to minimize them. In particular, do you think the result will be more or less precise than your pendulum experiment?

(b) Perform the experiment, taking at least ten independent measurements, and report the data and results with uncertainty.

Solution. Our formula for g is

$$g = \frac{2\ell(1+\beta)}{t^2\sin(\theta)}$$

where $\beta = I/MR^2$ of the rolling object, and ℓ, t are the distance and time for the path. Let's assume you found a nice object, like a hollow soup can with its ends removed, so that β is known relatively precisely. Then the uncertainty is

$$\Delta g = g\sqrt{\left(\frac{2\Delta t}{t}\right)^2 + \left(\frac{\Delta \ell}{\ell}\right)^2 + \left(\frac{\cos(\theta)\Delta\theta}{\sin(\theta)}\right)^2}.$$

Given the above, you definitely want a ramp as long as possible, and there's a tradeoff with the angle: if the angle is very large, t will be small so that the relative error on t will be large, while if the angle is very small, the relative error on θ will be large. So in practice you want to choose a moderately small, but not too small value of θ.

Some reasonable ballpark numbers are $\theta = (10 \pm 1)^\circ$, and $t = (3 \pm 0.3)\,\text{s}$, so you probably can't easily get an uncertainty smaller than a few percent. The overall result will be less precise than the pendulum experiment, because for the pendulum there is no $\Delta\theta$ term, and you can measure $N \gg 1$ periods in a single trial so that the relative error on t falls as $1/N$. With the ramp, you can partially compensate by doing N separate trials, so that the relative error at best falls as $1/\sqrt{N}$, which isn't as good. It might not even be as good as $1/\sqrt{N}$, because your uncertainties may not be independent: you might systematically overestimate or underestimate the time or angle.

[3] **Problem 17.** [A] Consider N independent measurements of the same quantity, with results $x_i \pm \Delta x_i$. They can be combined into a single result by taking a weighted average. What is the optimal weighted average, which minimizes the uncertainty, and what is the corresponding uncertainty?

Solution. Let the weights be w_i, so we report the value

$$\overline{x} = \sum_i w_i x_i.$$

The uncertainty obeys

$$(\Delta\overline{x})^2 = \sum_i w_i^2 (\Delta x_i)^2.$$

A tempting but incorrect way to minimize this quantity is to set the derivative with respect to w_i equal to zero. This doesn't work because the solution is just $w_1 = \ldots = w_N = 0$, which isn't a weighted average at all. To actually have a weighted average, we need the weights to sum to one,

$$\sum_i w_i = 1.$$

This is an optimization problem with a constraint, which can be solved with Lagrange multipliers.

However, for this particular problem, the constraint is simple enough to handle manually. Because of the constraint, if one increases some weight, then one must decrease others. At the minimum, the effect of increasing any weight infinitesimally and decreasing another by the same amount must be zero, as if it weren't, we could just adjust those two weights to get a lower uncertainty. Setting the change in the uncertainty due to adjusting w_i and w_j in this way to zero gives

$$0 = d(w_i^2)(\Delta x_i)^2 + d(w_j^2)(\Delta x_j)^2 = (2\,dw)(-w_i(\Delta x_i)^2 + w_j(\Delta x_j)^2).$$

This tells us that $w_i \propto 1/\Delta x_i^2$, which means

$$w_i = \frac{1/(\Delta x_i)^2}{\sum_j 1/(\Delta x_j)^2}.$$

Note that all measurements are included in the optimal average, no matter how bad they may be. The corresponding uncertainty is

$$\Delta x = \left(\sum_i \frac{1}{(\Delta x_i)^2} \right)^{-1/2}.$$

All of the examples above involve combining continuous quantities, so we'll close this section with some applications to "counting" experiments, which work slightly differently.

Example 4

A fair coin is tossed 1000 times, and the number of heads is counted. If this process is repeated many times, what is the standard deviation of the number of heads?

Solution

Consider one trial of 1000 tosses. The number of heads is $X = X_1 + X_2 + \ldots + X_{1000}$, where

$$X_i = \begin{cases} 1 & \text{heads on toss } i, \\ 0 & \text{tails on toss } i. \end{cases}$$

Of course, the mean of each of these variables is $\langle X_i \rangle = 0.5$, so that the mean of X is 500. In addition, the X_i are independent of each other, so the variances add. The variance of each one of them is

$$\operatorname{var} X_i = \langle X_i^2 \rangle - \langle X_i \rangle^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Thus, the standard deviation of the number of heads is

$$\sqrt{\operatorname{var} X} = \sqrt{1000/4} \approx 16.$$

So getting 520 heads would not be surprising, but if you got 550, you might be justified in suspecting the coin isn't fair. (Also, the number of heads is very close to normally distributed, by the central limit theorem mentioned above.) To check whether you understand this, you can redo it with a general probability p of getting heads, where you should get $\sqrt{1000\,p(1-p)}$.

[3] **Problem 18.** At any moment, a Geiger counter can click, indicating that it has detected a particle of radiation. Suppose that there is an independent probability $\alpha\,dt$ of clicking at each infinitesimal time interval dt. Let the number of clicks observed in a total time T be X.

(a) Find the expected value and standard deviation of X, and thereby compute its relative uncertainty. (Hint: split the total time into many tiny time intervals, and let X_i be the number of clicks in interval i, so $X = \sum_i X_i$.)

(b) Using a Geiger counter on a sample, you hear 197 clicks in 5 minutes of operation. Estimate the activity α of the sample (i.e. the expected clicks per second), with uncertainty. If you measure for longer, how does the uncertainty reduce over time?

(c) Now suppose that for a different sample, $X = 0$ after 5 minutes. Estimate the activity α of the sample (i.e. the expected clicks per second), with a reasonable uncertainty. If you measure for longer, and continue to hear no clicks, how does the uncertainty reduce over time?

Solution. (a) There are $N = T/dt$ time intervals. Using the hint and applying linearity of expectation,

$$\langle X \rangle = \sum_i \langle X_i \rangle = N(\alpha\,dt) = \alpha T.$$

Since the X_i are independent, their variances add. The variance of X_i is

$$\langle X_i^2 \rangle - \langle X_i \rangle^2 = \alpha\,dt - (\alpha\,dt)^2 \simeq \alpha\,dt.$$

Thus, by adding the variances, we have

$$\operatorname{var} X = \alpha T$$

so the standard deviation is $\Delta X = \sqrt{\alpha T}$. The relative uncertainty is $\Delta X/\langle X\rangle = 1/\sqrt{\alpha T}$.

(b) Applying the formulas above, we estimate

$$\alpha = \frac{197}{T} = 0.66\,\mathrm{s}^{-1}$$

with an uncertainty of

$$\Delta\alpha = \frac{\alpha}{\sqrt{\alpha T}} = \sqrt{\frac{\alpha}{T}} = 0.05\,\mathrm{s}^{-1}.$$

The uncertainty falls as $1/\sqrt{T}$. Note that this is very similar to previous results we've found, where the uncertainty falls as $1/\sqrt{n}$ where n is the number of trials. In some sense, each instant of time we wait is another trial here.

(c) Of course, we estimate $\alpha = 0$, but then the formulas above imply $\Delta\alpha = 0$ and hence that we are absolutely certain $\alpha = 0$, which is absurd. (If you don't think that's absurd, note that the same result would have occurred if we had heard zero clicks in an *arbitrarily short* time interval, such as a nanosecond.)

This is a case where the basic rules of uncertainty propagation break down, and we need to think. The point of giving an uncertainty is to indicate the range of parameter values compatible with the data we observed. Now, the probability of having no clicks in time T is $e^{-\alpha T}$. If $\alpha T \gg 1$, then it would be very unlikely to have no clicks, so we can rule out $\alpha \gg 1/T$. But if $\alpha T \lesssim 1$, this isn't unlikely at all. Thus, your uncertainty window should be $\alpha \in [0, c/T]$ where c is an order-one number, whose value depends on the specific statistical procedure you use. (Note that the upper bound falls as $1/T$, not $1/\sqrt{T}$.)

[4] **Problem 19.** [A] This problem extends problem 18 to derive some canonical results.

(a) Let $\lambda = \alpha T$. Find the probability $p(X = k)$ of hearing exactly k clicks in terms of λ and k.

(b) To check your result, show that the sum of the $p(X = k)$ is equal to one.

(c) ⋆ In the limit $\lambda \gg 1$, show that the probabilities $p(X = k)$ approach those of a normal distribution with the mean and standard deviation calculated in problem 18, thereby providing an example of the central limit theorem at work. This is a rather involved calculation, which will use many of the techniques from **P1**. It will also require Stirling's approximation,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

for $n \gg 1$, which we mentioned in **P1**. (Hint: because the relative uncertainty falls as λ increases, start by writing $k = \lambda(1+\delta)$ for $|\delta| \ll 1$, and expand in powers of δ. Be careful not to drop too many terms, as δ is small, but $\lambda\delta$ isn't.)

Solution. (a) Following the notation of problem 18, we have $X = \sum_i X_i$, and we get k clicks if precisely k of the X_i are equal to 1. Thus,

$$p(X = k) = \binom{N}{k}(\alpha\,dt)^k\,(1-\alpha\,dt)^{N-k} \approx \frac{N^k}{k!}(\alpha\,dt)^k\,(1-\alpha\,dt)^N = \frac{\lambda^k}{k!}e^{-\lambda}.$$

This is known as the Poisson distribution.

(b) This follows from the Taylor series of the exponential,

$$\sum_{k=0}^{\infty} p(X=k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = 1.$$

(c) Using Stirling's approximation, we have

$$\begin{aligned} p(X=k) &= \frac{1}{\sqrt{2\pi k}} \left(\frac{\lambda e}{k}\right)^k e^{-\lambda} \\ &= \frac{1}{\sqrt{2\pi\lambda(1+\delta)}} \left(\frac{e}{1+\delta}\right)^{\lambda(1+\delta)} e^{-\lambda} \\ &\approx \frac{1}{\sqrt{2\pi\lambda}} e^{\delta\lambda} (1+\delta)^{-\lambda(1+\delta)} \end{aligned}$$

where we used the fact that $\delta \ll 1$.

Now we need to use a technique from **P1**. Letting the final term be equal to $1/y$, we have

$$\log y = \lambda(1+\delta)\log(1+\delta) = \lambda(1+\delta)\left(\delta - \frac{\delta^2}{2} + \mathcal{O}(\delta^3)\right) = \delta\lambda + \frac{\delta^2\lambda}{2} + \mathcal{O}(\delta^3\lambda).$$

In **P1**, we only expanded up to the first term, but here we need to keep the order δ^2 term. The reason is we want an approximation that works for the whole peak of the probability distribution, and we know it has relative uncertainty $1/\sqrt{\lambda}$, which means we need to take $\delta \sim 1/\sqrt{\lambda}$. That implies that $\delta^2\lambda$ is of order one and cannot be dropped, but $\delta^3\lambda$ is small and can be dropped. Anyway, plugging this in, we find

$$p(X=k) \approx \frac{1}{\sqrt{2\pi\lambda}} e^{-\delta^2\lambda/2} = \frac{1}{\sqrt{2\pi\lambda}} e^{-(k-\lambda)^2/2\lambda}$$

which is precisely a normal distribution with the appropriate mean and standard deviation.

[3] **Problem 20.** IPhO 2023, problem 1, parts A, B, and D.3. A short derivation of the key features of Brownian motion. It requires only the ideas of this problem set, and some basic mechanics.

4 Data Analysis

Idea 5

All data analysis for the USAPhO and IPhO can be done using extremely basic methods. Sometimes, it suffices to just calculate a value based on a single data point, or by cleverly using a pair of data points. When this isn't enough, you'll have to do graphical data analysis, which will usually correspond to drawing a line and measuring its slope and intercept. This is quite limited compared to modern statistical tools, but also can be surprisingly powerful.

Example 5

The activity of a radioactive substance obeys $A(t) = A_0 e^{-t/\tau}$. Using measurements of t and $A(t)$, plot a line to find A_0 and τ.

Solution

To handle exponential relationships, take the logarithm of both sides to get

$$\log A(t) = \log A_0 - t/\tau.$$

Then a plot of $\log A(t)$ vs. t has slope $-1/\tau$ and y-intercept $\log A_0$.

[1] **Problem 21.** For a power law $y = \alpha x^n$ where y and x are measured, what line can be plotted to find α and n?

Solution. We have

$$\log(y) = \log(\alpha x^n) = \log(\alpha) + n \log(x).$$

Thus, if we plot $\log y$ against $\log x$, the slope will be n and the y-intercept will be $\log(\alpha)$.

[2] **Problem 22.** The rate R of electron emission from a solid in an electric field E is

$$R = \beta e^{-E_0/E}$$

for some constants β and E_0. The exponential form is because the effect is due to quantum tunneling, as will be discussed in **X2**.

(a) If E and R are measured, what line can be plotted to find β and E_0? Make sure the dimensions of your equations make sense.

(b) Suppose both β and E_0 have 1% uncertainty. For small E, which is more important for the uncertainty of R? What about for large E? Around where is the crossover point?

Solution. (a) Naively, if we take the natural log of the equation, we get

$$\log R = -\frac{E_0}{E} + \log \beta.$$

Plotting $\log R$ versus $1/E$ gives a line with slope $-E_0$ and y-intercept $\log \beta$.

But the dimensions of this equation look wrong, because R and β are dimensionful quantities. Technically, we are really plotting dimensionless quantities, involving the ratios of E and R to a standard unit of electric field and rate, respectively. For example, in SI units, we would have

$$\log \frac{R}{1\,\text{Hz}} = -\frac{E_0/(1\,\text{V/m})}{E/(1\,\text{V/m})} + \log \frac{\beta}{1\,\text{Hz}}.$$

On the page, we literally plot $\log(R/(1\,\text{Hz}))$ versus $(1\,\text{V/m})/E$. This gives a line with dimensionless slope $-E_0/(1\,\text{V/m})$ and intercept $\log(\beta/(1\,\text{Hz}))$. You don't have to keep this technicality in mind, but this sort of thing is always implicitly happening whenever one plots data.

(b) The uncertainty in β alone always gives a 1% uncertainty in R. By the tangent line approximation, the relative uncertainty in R due to E_0 is $\Delta E_0/E = (E_0/E) \times 1\%$. Therefore, the uncertainty in E_0 dominates for $E \ll E_0$, while that in β dominates for $E \gg E_0$. In the latter limit, $e^{-E_0/E} \approx 1 - E_0/E$. The crossover is around $E \sim E_0$.

Example 6

Suppose that y and x are related nonlinearly, as

$$y = bx + ax^2.$$

For example, this could model the force due to a non-Hookean spring. Using measurements of x and y, plot a line to find a and b.

Solution

If we divide by x, we find

$$\frac{y}{x} = ax + b.$$

Therefore, we can plot y/x versus x, which gives a line with slope a and intercept b. More generally, we can plot a line whenever we can rearrange a given relation into the form

$$(\text{known}) = (\text{unknown})(\text{known}) + (\text{unknown})$$

where all four terms can be arbitrarily complicated. In this way, it is possible to turn a lot of very nonlinear relations into lines.

[3] **Problem 23.** Some more examples of finding lines to plot.

(a) Suppose that you are given points (x, y) that lie on a circle centered at $(a, 0)$ with radius r. What line can be plotted to find a and r?

(b) Consider an Atwood's machine with masses m and $M > m$. The acceleration of the machine is measured as a function of M. However, since the pulley has mass, it slows the acceleration of the Atwood's machine, so that

$$a = \frac{M - m}{M + m + \delta m} g.$$

Find a line that can be plotted to find g and δm, assuming m, M, and a are known. This is an example of how plotting a line can separate out a systematic error, i.e. the value of δm, which would be impossible if only one value of M were used.

(c) Suppose an object is undergoing simple harmonic motion with amplitude A and angular frequency ω. Given measurements of the position x and velocity v, what line can be plotted to find A and ω?

Solution. (a) The equation of the circle is

$$(x - a)^2 + y^2 = r^2, \quad y^2 + x^2 = 2ax + r^2 - a^2$$

Plotting $y^2 + x^2$ vs. x will give a slope of $2a$ and a y-intercept of $r^2 - a^2$. Combining the two pieces of information yields a and r.

(b) The equation can be slightly rearranged to give

$$\frac{M-m}{a} = \frac{M+m}{g} + \frac{\delta m}{g}.$$

Therefore, a plot of $(M-m)/a$ vs. $M+m$ has slope $1/g$ and y-intercept $\delta m/g$.

(c) By conservation of energy, $A^2 = x^2 + v^2/\omega^2$, so

$$x^2 = A^2 - v^2/\omega^2.$$

Thus, a plot of x^2 vs. v^2 has y-intercept A^2 and slope $-1/\omega^2$.

[3] **Problem 24.** USAPhO 2012, problem A2. (This one requires basic thermodynamics.)

[3] **Problem 25.** USAPhO 2011, problem A2.

[3] **Problem 26.** INPhO 2018, problem 7. (This one requires basic fluid dynamics.)

[3] **Problem 27** (USAPhO 2024)**.** An experimentalist drives a series RLC circuit with a sinusoidal voltage $V(t) = V_0 \cos \omega t$. In **E6**, you will learn how to show that the voltage across the capacitor, in the steady state, oscillates with amplitude

$$V_c = \frac{V_0}{\sqrt{(1-\omega^2/\omega_0^2)^2 + (\omega/\omega_0 Q)^2}}$$

where ω_0 is the resonant angular frequency and Q is the circuit's quality factor. The experimentalist takes the following data near the resonance, for a fixed value of V_0:

ω (rad/s)	133.0	133.5	134.0	134.5	135.0	135.5	136.0	136.5	137.0
V_c (Volts)	3.64	4.76	6.52	8.53	8.18	6.06	4.44	3.42	2.75

Find the values of ω_0 and Q as accurately as possible. Uncertainty analysis is not required. (Hint: you don't always have to draw a line. Carefully examine and approximate the given formula.)

Solution. See the official solutions to problem B3, part a.b.ii. A good answer for ω_0 is within the range (134.7 ± 0.1) rad/s, and a good answer for Q is within 88 ± 3.

5 Estimation

Estimation is a useful skill for checking the answers to real-world problems.

Example 7

Estimate the circumference of the Earth.

Solution

If you know that the United States is 3,000 miles wide, and there is a time zone difference of three hours between California and New York, then a reasonable estimate is 24,000 miles. Or, if you know the factoid that light can go about seven times around the Earth in a second,

then a reasonable estimate is $(3/7) \times 10^8\,\text{m} \approx 4 \times 10^7\,\text{m}$.

Let's check these results are compatible. There are about 5 miles in 8 kilometers, a fact you can get by remembering how your car's speedometer looks, or by noting that 3 feet are about 1 meter. Then $4 \times 10^4\,\text{km} \approx (5/8) \times 4 \times 10^4\,\text{mi} = 2.5 \times 10^4\,\text{mi}$, so the two results are compatible. There are probably at least a hundred more ways to perform this estimation.

Example 8

Estimate the density of air, and compare this to the density of water.

Solution

We can directly use the ideal gas law, $PV = nRT$. The density is $\rho = \mu n/V$ where μ is the mass of one mole of air, so

$$\rho = \frac{\mu P}{RT}.$$

Atmospheric pressure is about $10^5\,\text{Pa}$, typical temperatures are about $300\,\text{K}$, and air is mostly N_2, which has a molar mass of $\mu = 28\,\text{g/mol}$, so

$$\rho = \frac{(0.028)(10^5)}{(8.3)(300)} \frac{\text{kg}}{\text{m}^3} \approx 1\,\frac{\text{kg}}{\text{m}^3}.$$

The density of water is, almost by definition,

$$\rho_w \approx 10^3\,\frac{\text{kg}}{\text{m}^3}.$$

Most liquids and solids have densities within an order of magnitude of this, since in all cases the atoms are packed close together. Evidently, air molecules are about a factor of $(10^3)^{1/3} = 10$ times further apart than typical water molecules.

Example 9

Estimate how much useful power you can produce in a short burst.

Solution

A typical person weighs about $75\,\text{kg}$ and can run up a $3\,\text{m}$ high staircase in around $3\,\text{s}$, so

$$P = mgv = (75)(10)(3/3)\,\text{W} \approx 750\,\text{W}.$$

This happens to be about equal to one horsepower. Typical steady state power outputs are several times smaller, and the numbers for elite athletes are several times larger.

Remark

It's important not to overthink estimation questions. If you already know what answer you're targeting, you can always add ad hoc factors to get closer, but that's not a good instinct in novel situations, where you don't know the answer. Instead, it's best to keep things simple, and only try to get within one or two orders of magnitude. Accordingly, the solutions to the problems below will also use very simple estimates, accurate to only an order of magnitude.

[2] **Problem 28.** Estimate the radius of the largest asteroid you could jump off of, and never return.

Solution. The escape velocity is $v = \sqrt{2GM/R}$, and we will assume a uniform spherical asteroid with density ρ. Rock is a few times denser than water, so $\rho \approx 3 \times 10^3$ kg/m^3 and $M \approx \frac{4}{3}\pi\rho R^3$. Humans can jump around half a meter, which determines $v = \sqrt{2gh}$. Thus

$$2gh = \frac{2G}{R}\frac{4}{3}\pi\rho R^3.$$

Plugging in numbers gives $R \approx 2\,\text{km}$.

[3] **Problem 29.** Some questions about light energy.

(a) Estimate the number of photons emitted per second by a standard light bulb. (The energy of a photon is $E = hf$, and the frequency of a photon is related to the wavelength by $c = f\lambda$.)

(b) The Sun supplies power at an intensity of $1400\,\text{W/m}^2$ to the Earth. The nearest star is about 4 light years away. Assuming this star is similar to the Sun, about how many of its photons enter your eye per second?

Solution. (a) A standard light bulb has around 50 W of power. The power $P = NE$ where N is the number of photons emitted per second, and the wavelength range of visible light is 400 to 700 nm. Using 500 nm, we have

$$N = \frac{P\lambda}{hc} \sim 10^{20}\,\text{photons/s}$$

(b) The distance to the Sun is $1\,\text{AU} \sim 1.5 \times 10^{11}\,\text{m}$, which you can also reconstruct with Kepler's laws. 1 light year is $c(1\,\text{year}) \sim 10^{16}\,\text{m}$. Then the intensity from the star is reduced by a factor of $(1\,\text{AU}/4\,\text{ly})^2$ due to the inverse square law, so $I \sim 10^{-8}$ W/m^2.

The area of a human pupil depends on the light conditions, but is roughly $\pi r^2 = \pi(5\,\text{mm})^2$. Then the number of photons that enter it per second is $P\lambda/hc$, which gives $N \sim 10^6$ photons/s. That's plenty, so it's very easy to see such a star at night, while it might be difficult during the day because of the background light from the Sun.

[3] **Problem 30.** In this problem we build some intuition for energy. Humans require about 2000 Calories per day. One Calorie is about 4.2 kJ, enough energy to raise the temperature of 1 kg of water by 1 K. The most energy-dense foods are pure fat, carrying about 9 Calories per gram.

(a) Estimate the rate at which your body burns energy when at rest.

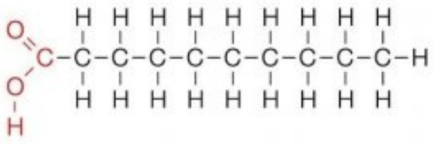
(b) Estimate the rate at which a human being radiates energy. (The Stefan–Boltzmann law states that the radiation power per unit area from a blackbody is σT^4, where $\sigma = 5.7 \times 10^{-8}\,\text{W/m}^2\text{K}^4$.) Compare to your answer in part (a).

(c) How many Calories per day would you "burn" if you only drank ice-cold water?

(d) The energy density of coal or oil is comparable to fat, since in both cases one is breaking similar chemical bonds. How much coal does it take to keep a 100 W light bulb on for a year?

We can estimate the energy density of fat from first principles. The binding energy of an electron in hydrogen is $\sim 10\,\text{eV}$, and strong chemical bond energies are roughly a factor of 10 smaller, since they roughly correspond to differences of nearby binding energies.

(e) Fats are digested by inputting energy to break the bonds in the molecules, then harvesting energy by combining the atoms into CO_2 and H_2O, which have somewhat more stable bonds.



Roughly estimate the energy content of a gram of fat, and compare to the true result.

Solution. (a) The power is

$$\frac{(2000\,\text{Cal})(4000\,\text{J/Cal})}{86400\,\text{s}} \approx 100\,\text{W}.$$

This is used to maintain homeostasis in your body, and is eventually dissipated as heat.

(b) A rough estimate for the surface area of the human body is $A \approx 1\,\text{m}^2$. Human skin is on the order of 300 K, so $P \sim A\sigma T^4 \approx 500$ W. This is wrong, because it can't be higher than (b). The point is that the radiation output by the human body is almost completely cancelled by the radiation input by the environment, which is at almost the same temperature (in absolute terms). For example, in typical indoor conditions, the environment is at 70° F and human skin is at 90° F, for a difference of about 10 K. So the power is smaller by a factor of $1 - (290/300)^4 = 0.13$, giving a reasonable 65 W. It's still a significant contribution, but not unreasonably large. Of course, in practice it's further reduced by wearing clothes, which trap a layer of insulating air and cloth near you.

(c) If you drink eight cups of water a day, that's 2 kg of water. Warming it from 0° C to a body temperature of around 35° C takes about 70 Calories.

(d) This is the same power we found in part (a), and a human's daily food intake corresponds to 2000/9 grams of fat per day. This corresponds to 80 kg of coal in a year. In fact, it's more than that, since coal power plants are far from 100% efficient. Alternatively, you can use one solar panel. (For comparison, a window-mounted air conditioner uses $\sim 1\,\text{kW}$, and a car on the highway uses $\sim 10\,\text{kW}$.)

(e) Most of the mass of the molecule shown is carbon, so let's count only the carbon, and say that the net energy released per carbon atom is $\sim 1\,\text{eV}$. Since the molar mass of carbon is 12 g, the energy per gram is

$$(1\,\text{eV})\frac{N_A}{12} \sim 8\,\text{kJ} = 2\,\text{Cal}$$

which isn't bad for such a naive estimate.

[2] **Problem 31** (Povey)**.** When human beings lose weight, most of it is by exhalation of carbon. About 20% of the air in the atmosphere is oxygen. When we breathe in and then out, about 25% of the oxygen is converted to carbon dioxide.

(a) Estimate the mass of air contained in a single breath.

(b) Estimate the amount of weight we lose every day by breathing alone.

Solution. (a) If you don't take a deep breath, you can barely blow up a crushed plastic water bottle with volume 0.5 L, so we can take this to be the volume of a breath.

From the ideal gas law $V/n = RT/P$, a mole of gas takes up 22.4 L of volume at atmospheric pressure and $T = 273\,\text{K}$, and we round this up to 25 L since body temperature is a bit warmer.

Air is mostly nitrogen (N_2) with molecular mass 28 g/mol. Then one breath should have a mass of $(0.5\,\text{L})/(25\,\text{L/mol})(28\,\text{g/mol}) \approx 0.6\,\text{g}$.

(b) Humans breathe around 10 to 15 times a minute, so let's use 12.5, giving around 20,000 breaths in a day, corresponding to 12 kg of air.

In each breath, 20% of the air is oxygen, and 25% of the oxygen is converted to carbon dioxide, for a net fraction of 5%. Carbon dioxide (CO_2) has a molecular mass of 44 g/mol, and oxygen is 32 g/mol. Thus we lose a proportion of $(44/32 - 1) = 0.375$ of the mass of the air we breathe in every day, corresponding to $(0.375)(0.05)(12\,\text{kg}) = 0.2$ kg. Most of the (non-water) mass of the food we eat leaves this way.

[2] **Problem 32.** How long a line can you write with a pencil? (Hint: graphene, a single layer of carbon atoms in a hexagonal lattice, absorbs about 2% of visible light.)

Solution. Pencil lead is made of graphite, a form of carbon. The spacing between the carbon atoms is $d \sim 10^{-10}\,\text{m}$, the length of the lead is $L \sim 10\,\text{cm}$, and the radius of the lead is $r \sim 1\,\text{mm}$. If the line is just thick enough to be opaque, then it will be $N \sim 10^2$ atoms thick. This yields a line of length

$$\frac{r^2 L}{Nrd} \sim 10^4\,\text{m}.$$

In actual research labs, graphene is often made by peeling Scotch tape off pencil markings.